# Eyal Weiss

*Curriculum Vitae*

Postdoctoral Researcher, Robotics Institute, Carnegie Mellon University

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## Summary


Postdoctoral researcher in AI and robotics working on heuristic search, automated planning and motion planning, with a focus on planning with uncertain or expensive-to-compute action costs and on combinatorial optimization for multi-agent systems. 5 journal and 10 conference papers; SoCS 2025 Best Paper Award; Adams Fellow.


## Research Interests

Artificial intelligence, robotics, heuristic search, automated planning, motion planning, multi-agent planning, combinatorial optimization; background in control theory and dynamical systems.

## Research Positions

**Carnegie Mellon University**, Pittsburgh, PA, USA — Aug 2026 – present
*Postdoctoral Researcher*, Robotics Institute

- Topics: AI planning, robotics, heuristic search, motion planning, combinatorial optimization
- Hosts: Prof. Maxim Likhachev (Search-based Planning Laboratory) and Prof. Jiaoyang Li (ARCS Lab)

**Technion – Israel Institute of Technology**, Haifa, Israel — Oct 2024 – Aug 2026
*Postdoctoral Scholar*, Faculty of Computer Science

- Topics: AI planning, robotics, heuristic search, motion planning
- Host: Prof. Oren Salzman (Computational Robotics Lab)

**Tel Aviv University**, Tel Aviv, Israel — Apr 2018 – Sep 2019
*Research Assistant*, School of Electrical Engineering

- Topics: control theory, dynamical systems, Boolean networks
- Supervisor: Prof. Michael Margaliot

## Education

**Bar-Ilan University**, Ramat Gan, Israel — Oct 2019 – Sep 2024
*Ph.D.*, Computer Science

- Thesis: *Graph Search and Automated Planning with Multiple Cost Estimators*
- Advisor: Prof. Gal A. Kaminka

**Tel Aviv University**, Tel Aviv, Israel — Mar 2018
*M.Sc.*, Electrical Engineering, *Cum Laude*

- Thesis: *Minimal Controllability and Minimal Observability of Conjunctive Boolean Networks*
- Advisor: Prof. Michael Margaliot

*B.Sc.*, Electrical Engineering, Tel Aviv University, *Summa Cum Laude* — Oct 2016

## Honours and Awards

- **Best Paper Award**, International Symposium on Combinatorial Search (SoCS) 2025
- Adams Fellowship (Israel Academy of Sciences and Humanities) for outstanding doctoral students 2022 – 2024
- Excellence Scholarship for research students, Computer Science Department, Bar-Ilan University 2023
- Milgat Hanasi (President's Scholarship) for excellent Ph.D. students 2021 – 2023
- Shlomo Shmeltzer Institute for Smart Transportation Scholarship 2017
- Excellent Student Prize for M.Sc. students, Faculty of Engineering, Tel Aviv University 2017
- Velger Prize for excellence in graduate studies in Systems and Control 2017
- Final Prize for excellent students in Electrical Engineering 2015
- Freescale Scholarship for excellent students in Electrical Engineering 2014
- Dean's Honors, every year of the B.Sc. 2012 – 2016

## Preprints

1. H. Peer, **E. Weiss**, R. Alterovitz and O. Salzman, "Generalizing Multi-Objective Search via Objective-Aggregation Functions", *arXiv preprint* arXiv:2509.22085, 2025.

## Journal Publications

1. **E. Weiss** and M. Margaliot, "A Generalization of Linear Positive Systems with Applications to Nonlinear Systems: Invariant Sets and the Poincaré–Bendixson Property", *Automatica*, 123, 109358, 2021.
2. **E. Weiss** and M. Margaliot, "Is My System of ODEs k-Cooperative?", *IEEE Control Systems Letters*, 5(1):73–78, 2021.
3. **E. Weiss** and M. Margaliot, "A Polynomial-Time Algorithm for Solving the Minimal Observability Problem in Conjunctive Boolean Networks", *IEEE Transactions on Automatic Control*, 64(7):2727–2736, 2019.
4. **E. Weiss** and M. Margaliot, "Output Selection and Observer Design for Boolean Control Networks: A Sub-Optimal Polynomial-Complexity Algorithm", *IEEE Control Systems Letters*, 3(1):210–215, 2019.
5. **E. Weiss**, M. Margaliot and G. Even, "Minimal Controllability of Conjunctive Boolean Networks is NP-Complete", *Automatica*, 92:56–62, 2018.

## Conference Publications

1. Y. Wang, **E. Weiss**, B. Mu and O. Salzman, "Bidirectional Search while Ensuring Meet-In-The-Middle via Effective and Efficient-to-Compute Termination Conditions", *Proc. International Joint Conference on Artificial Intelligence (IJCAI)*, 8987–8995, 2025.
2. Y. Sherma, **E. Weiss** and O. Salzman, "From Agent Centric to Obstacle Centric Planning: A Makespan-Optimal Algorithm for the Multi-Agent Warehouse Rearrangement Problem", *Proc. International Symposium on Combinatorial Search (SoCS)*, 136–144, 2025. **Best Paper Award.**
3. D. Gnad, L. Alon, **E. Weiss** and A. Shleyfman, "PDBs Go Numeric: Pattern-Database Heuristics for Simple Numeric Planning", *Proc. AAAI Conference on Artificial Intelligence*, 26507–26515, 2025.
4. **E. Weiss**, A. Felner and G. A. Kaminka, "Tightest Admissible Shortest Path", *Proc. International Conference on Automated Planning and Scheduling (ICAPS)*, 643–652, 2024.
5. **E. Weiss**, A. Felner and G. A. Kaminka, "A Generalization of the Shortest Path Problem to Graphs with Multiple Edge-Cost Estimates", *Proc. European Conference on Artificial Intelligence (ECAI)*, 2607–2614, 2023.
6. **E. Weiss** and G. A. Kaminka, "Planning with Multiple Action-Cost Estimates", *Proc. International Conference on Automated Planning and Scheduling (ICAPS)*, 427–437, 2023.
7. **E. Weiss** and M. Margaliot, "A Generalization of Linear Positive Systems", *27th Mediterranean Conference on*

*Control and Automation (MED)*, 340–345, 2019.

8. **E. Weiss** and M. Margaliot, "A Generalization of Smillie's Theorem on Strongly Cooperative Tridiagonal Systems", *IEEE Conference on Decision and Control (CDC)*, 3080–3085, 2018.
9. **E. Weiss** and M. Margaliot, "Observability Analysis and Observer Design for Boolean Control Networks: A Sub-Optimal Polynomial-Complexity Algorithm", *IEEE International Conference on the Science of Electrical Engineering (ICSEE)*, 2018.

## Workshop Publications

1. D. Gnad, L. Alon, **E. Weiss** and A. Shleyfman, "PDBs Go Numeric: Pattern-Database Heuristics for Simple Numeric Planning", *ICAPS'24 Workshop on Heuristics and Search for Domain-Independent Planning (HSDIP)*, 2024.
2. **E. Weiss**, A. Felner and G. A. Kaminka, "A Generalization of the Shortest Path Problem to Graphs with Multiple Edge-Cost Estimates", *ICAPS'23 Workshop on Reliable Data-Driven Planning and Scheduling (RDDPS)*, 2023.
3. **E. Weiss** and G. A. Kaminka, "Planning with Dynamically Estimated Action Costs", *ICAPS'22 Workshop on Reliable Data-Driven Planning and Scheduling (RDDPS)*, 2022.
4. **E. Weiss** and G. A. Kaminka, "Position Paper: Online Modeling for Offline Planning", *ICAPS'22 Workshop on Reliable Data-Driven Planning and Scheduling (RDDPS)*, 2022.

## Invited Talks

- Search-based Planning Laboratory (SBPL), Carnegie Mellon University — May 2025
- Computational Robotics Lab (CRL), Technion — Jun 2024
- Adams Fellowship Conference, Israel Academy of Sciences and Humanities, Jerusalem — Feb 2023
- Bar-Ilan Symposium on Foundations of AI (BISFAI), Ramat Gan — Feb 2023
- Israeli Association for Artificial Intelligence (IAAI) Symposium, Haifa — Jun 2022

## Software

- PlanDEM – domain-independent planner for dynamically estimated action models (author)
- MEET – bidirectional search (IJCAI 2025); being integrated into OMPL
- Warehouse Rearrangement – reference implementation (SoCS 2025)
- Numeric Fast Downward – numeric pattern-database heuristics (AAAI 2025)
- CSNA – cost-sensitive neighborhood aggregation for graph neural networks

## Service

**Reviewer, conferences:** ICAPS, AAAI, IJCAI, RSS, ICANN

**Reviewer, journals:** IEEE Robotics and Automation Letters; Automatica; IEEE Transactions on Automatic Control; European Journal of Control; IEEE Transactions on Network Science and Engineering; Optimization and Engineering

**Organizing committee**

- ICAPS Workshop on Reliability In Planning and Learning (RIPL) — 2026
- ICAPS Workshop on Heuristics and Search for Domain-independent Planning (HSDIP) — 2025
- ICAPS Workshop on Reliable Data-Driven Planning and Scheduling (RDDPS) — 2023, 2024

**Session chair**

- European Conference on Artificial Intelligence (ECAI) 2023
- IEEE Conference on Decision and Control (CDC) 2018

## Student Supervision and Mentoring

- Research mentoring of graduate students at CRL, Technion, leading to joint publications (IJCAI 2025, SoCS 2025) 2024 – 2026
- Supervision of Computer Science graduation projects in heuristic search, Technion 2025 – 2026
- Supervision of gifted high-school students' research projects, Alpha program, Bar-Ilan University 2022 – 2024
- Supervision of Electrical Engineering graduation projects in autonomous driving with mini-scale cars, Tel Aviv University 2018 – 2019

## Teaching Experience

**Bar-Ilan University**, Department of Computer Science – *Teaching Assistant* 2020 – 2024

- Artificial Intelligence (2024); Statistical Methods in Computer Science (2023); Introduction to Intelligent, Smart and Cognitive Systems (2021–2022); Object Oriented Programming (2021); Computer Structure (2020–2022)

**Tel Aviv University**, School of Electrical Engineering – *Teaching Assistant and Laboratory Instructor* 2016 – 2020

- Stochastic Processes (2019–2020); Signals and Systems (2017–2020); Random Signals and Noise (2017–2020); Optimal Control (2016–2017)
- Laboratory of Advanced Control (2016–2020)

## Professional Experience

**Mellanox Technologies** (now part of NVIDIA), Israel – *Network Architecture Intern* 2015

**Mellanox Technologies** – *Chip Design Intern* 2013 – 2014

## Skills and Languages

- Programming: C, C++, Python, Java, MATLAB, Bash; comfortable picking up additional languages as needed
- Laboratory: robotic systems, electronics and low-level controllers
- Languages: Hebrew (native), English (full professional proficiency)

## Community and Volunteering

- Tutor, economic empowerment program of "Misdar Dorshei Tov" 2018 – 2022
- Technical editor, "Good to Know" project, Tel Aviv University 2017
- Personal mentor for youth at risk, "Shimony" Children's Home 2012 – 2014